

Improving Fitness of Elite Handball Players:

Small-Sided Games vs. High-Intensity Intermittent Training

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ABSTRACT

The present study was designed to compare the effects of high-intensity intermittent training (HIIT) and small-sided games (SSGs training) on fitness variables of elite handball players. Eighteen highly trained players (mean age $\pm$ SD: 25.6 $\pm$ 0.5) were assigned to either HIIT or SSGs group training protocols twice per week for 8 weeks. The HIIT consisted of 12-24 x 15s high-intensity runs interspersed by 15s of recovery. The SSGs training consisted of 3 against 3 small-sided handball games. Both training methods were matched for exercise duration and recovery at each training session. Before and after training, the following fitness variables were assessed: speed – 10m and 20m sprint time, agility - handball agility specific test (HAST), upper arm strength – 1RM bench press test, lower limb power – counter-movement jump tests with (CMJarm) and without (CMJ) arm movement, and aerobic fitness (YYIRTL1). Significant improvement was found in the YYIRTL1 (23.3 and 26.3%, respectively), 10m (2.3 and 4.1%, respectively) and 20m (2.1 and 4%, respectively) sprint, HAST (1.1 and 2.2%, respectively), 1RM bench press (6.8 and 12.3%, respectively), CMJ (7.4 and 10.8%, respectively), and CMJarm (6.4 and 8.9%, respectively) following training in both groups ($p < 0.05$ for all). There was a significantly greater improvement in 10m and 20m sprint, HAST, 1RM, CMJ, and CMJarm following the SSGs training compared to the HIIT ($p < 0.05$ for all). These results indicated that both HIIT and SSGs are effective training methods for fitness development among elite adult handball players. However, SSGs training may be considered as the preferred training regimen for improving handball-specific fitness variables during the in-season period.

Key Words speed, power, performance, training, agility

INTRODUCTION

Handball is a strenuous contact team sport that involves high intensity short-duration activities such as sprinting, jumping, turning, pushing, blocking, and throwing (15,36,38). It therefore seems that the ability to continuously perform intermittent high-intensity actions throughout the game is crucial for handball players (33,38). When focusing on sport-specific characteristics and requirements, researchers in the field of sport performance, have proven that in addition to the technical and tactical demands, fitness and handball-specific skills are important factors for successful participation at the elite handball level (e.g., 16,25,28,29,36). Physiological responses during the handball match emphasize the significant involvement of both aerobic and anaerobic energy systems (9,25,26,33,38). Povoas et al. (33) reported that during a 60-min match, players covered about 4 km at a mean intensity of 87% of maximal heart rate, demonstrating that about 90% of the total energy is driven aerobically (26). Using video analysis, Cuesta (7) showed that players of the Spanish national team performed a mean of 190 rhythm variations, 279 changes of direction, and 16 jumps, for a total of 485 high-intensity movements with energy supplies mostly furnished by ATP-CP and anaerobic capacities.

The daily handball practice itself has a direct effect on most of the determinant qualities in the game. Nevertheless, performing additional handball-specific training based on strength, power, speed, agility, and high-intensity aerobic intermittent activity is highly recommended (5,15,27). Therefore, the development and improvement of these specific physical capacities appears to be of great interest to handball coaches and players. High-intensity intermittent training (HIIT) constitutes an effective method of improving some of these variables (3,5). For instance, it was found that HIIT enhances aerobic capacity (1,5) and the ability to perform high-intensity actions with high solicitation of the anaerobic system (3,14). The effectiveness of this

methodology is optimized since intensity can be individualized and controlled using a reference for the involved workload (1,40). The SSGs is a training method that has recently become the focus of scientific research because of its ability to develop physical capacities together with sport-specific tactical and technical skills (3,4,11,18,31,35). SSGs training have acquired great success, especially in soccer, where it offers many practical advantages that have led to its popularity as a training modality (18). The primary benefits of SSGs training are that the game can replicate the movement patterns, as well as the physiological demands and technical requirements of competitive match play (11,31), whilst also requiring players to make decisions under pressure and fatigue. In addition, compared with traditional fitness training sessions, SSGs training is thought to increase player compliance and motivation, since it is perceived to be sport-specific exercise that maximizes the training time spent with the ball (3). Previous researchers (e.g., 18,19,35) have documented the physiological responses and long-term training effects of SSGs training in soccer. However, currently there is very little data on the effectiveness of SSGs as a training method in handball (3,5). To our knowledge, only one study (3) has investigated the respective effects of specific versus generic training on the repeated sprint ability or intermittent endurance capacity of young handball players. This study demonstrated that small-sided handball games, in addition to the usual training, resulted in similar improvements in repeated sprint ability and aerobic power as did intermittent high-intensity run training (3). At present, no data are available regarding the long-term effects of SSGs training versus HIIT on the fitness of elite adult handball players during the in-season period. Such data may be of great importance to handball coaches and players, especially when assessing the players' abilities using specific tests that replicate handball-specific activities such as the shuttle run (38), short sprints (28), vertical jumps (2), maximal strength tests (27) and agility drills (39).

The purpose of the present study, therefore, was to compare HIIT to SSGs training in improving selected fitness variables of elite male handball players during the in-season period. We hypothesized that while both training methods will improve the players fitness, SSGs training will lead to greater improvement in handball-specific activities (agility, jumps, dynamic strength, and short sprints), and HIIT will lead to greater improvement in standard aerobic performance (YYIRL1).

METHODS

Experimental Approach to the Problem

For many years, traditional intermittent-type training served as a major tool by which team sport coaches developed relevant fitness variables of players. In recent years, however, coaches of team sports often use SSGs in their training programs due to their proposed advantages in terms of specificity of movements, inclusion of decision making, and the potential to develop multiple fitness components all in conjunction with technical skills and tactical aspects (18). The present study was designed to compare the effects of these two training methods on relevant fitness variables of elite handball players.

The study was conducted during the first part of the handball in-season period (October-November). Overall, the study lasted 10 weeks and consisted of one week of pre-testing, eight weeks of specific training, and one week of post-testing. Participants were randomly assigned to one of the two training groups – HIIT ($n = 9$) and SSG ($n = 9$) – after matching for aerobic (YYIRL1 results) and anaerobic (10m and 20m sprint times) fitness, and for specific role in the team. Each of these training protocols was performed twice a week. To isolate the effect of the

two training protocols, the additional fitness training sessions (e.g. technical, tactical, strength) during the 8 weeks of training were identical for both groups.

Pre-training and post-training aerobic (YYIRTL1), anaerobic (10m and 20m sprints, CMJ, CMJarm, HAST), and strength (1RM bench press) fitness characteristics were evaluated, since they replicate the activity patterns and the requirements of the game of handball (15,28,29,33,36). All testing and training sessions were performed at the same time of the day (5:00 PM to 7:00 PM) and in a similar ambient temperature (19-22 C°). Coaches and players were asked to avoid intense exercise on the day prior to the tests and to consume their usual meal at least 3 hours before the scheduled testing time. During all training sessions (SSGs and HIIT), heart rate (HR) was continuously recorded using a telemetry system (Hosand Technologies Srl, Verbania, Italy) with an electrode transmitter belt (T31, Polar, Electro, Kempele, Finland) fitted to the participant's chest.

Participants

Eighteen male elite handball players (age 25.6 ± 0.5 ; body height 187.3 ± 5.4 cm; body mass 90.5 ± 9.1 kg; body fat % 12.7 ± 3.2), members of a top team of the national Israeli league, volunteered to participate in the study. All players had at least five years of competitive experience at the national or the international level. Seven of the players were included in the national handball team squad at the time of data collection. The players trained five days per week and competed during the weekend. All players were healthy and were not taking any medication. The study was approved by the Research Ethics Committee of the Zinman College at the Wingate Institute, Israel. The participants were fully informed about the procedures to be used and gave voluntary written consent to participate.

Training Protocols

Training started one week after the baseline testing and consisted of two sessions per week (on Sunday and Wednesday) of either HIIT or SSGs performed over a period of eight consecutive weeks. Both training programs were structured according to a gradual progress plan designed to maximize final performance (Table 1). The HIIT program consisted of 12-16min intermittent running of 15s activity over 40-m shuttles interspersed by 15s of passive walking recovery. An example of the intermittent runs pattern is presented in Figure 1. The choice of the running velocity was justified by the fact that this kind of protocol allows participants to elicit and maintain VO_2max (4,14,15). Exercise intensity (controlled by the intermittent running distance covered over 15s) was set at 90-95% of the maximal aerobic shuttle run field test final speed, which was shown to induce an intensity that corresponds to $\sim 120\%$ of VO_2max (14,21). In order to ensure the appropriate running intensity, an audio signal, by whistle, was provided every 15 sec. The SSGs training program consisted of five 2.25-3.10 min bouts of small handball games with a passive recovery of 1 min between bouts (Table 1). SSGs were organized in 3-a-side teams excluding goalkeepers on a playing court delimited by an area of 20 x 20 m (half regular handball court) wherein the players had to score in two mini-goals (1.2 x 0.9 m). Some playing rules were created in order to avoid game breaks, thus ensuring continuity of the games and consequently keeping high exercise intensity. For instance, walking and dribbling were not allowed, defense stops as for those achieved by regular fouls were sanctioned with ball turnover, the maximal time to complete an attack before losing ball possession was preset at 20s. Although players did not receive any specific instructions regarding the intensity level, they were asked to maintain high pace throughout each of the games. HIITs and SSGs were matched for activity and recovery time at each training session. In addition, to isolate the effect of the current two training

protocols, other fitness training sessions (e.g., technical, tactical, and strength) were conducted identically in both groups during the 8 weeks of the study. Specifically, the average total training time for each group was 11.5 hours per week, including similar technical, tactical strength, and basic skills' drills. Strength training included two sessions/week of upper and lower limb exercises (Bench Press, Shoulder Press, Pullover, Half Squat, Deadlift, Lunge), with an average load of 75% of 1RM, 8-10 repetitions, 3 sets for each exercise.

TABLE 1 ABOUT HERE

FIGURE 1 ABOUT HERE

Testing Procedures

Anthropometric variables of height (m), body mass (kg), and body fat (%) were measured for each participant. Height and body mass measurements were made on a leveled platform scale (SECA model 284, Germany) with an accuracy of 0.001 m and 0.05 kg, respectively. Percentage of body fat was calculated from measurements of seven skinfold thickness (21). Participants performed two similar sets of tests before and after the eight-week training period. The first set of tests was conducted on two non-consecutive days during the week prior to the beginning of the training program. The second set of tests was conducted under the same conditions during the week after completion of the training program. Both sets of tests were performed using the same procedures, and with the same technician, who was blind to the training-group affiliation. All participants were familiar with the testing procedures, as they had performed them on previous occasions. Before each set of tests, the participants performed a standard warm-up that included 8 min of jogging, followed by 10 min of dynamic stretching exercises, two 20-m sprints and jumping drills. About 20-30 min separated the different tests on each day. All running and jumping tests were performed at the same location and under similar environmental conditions as

in the training sessions. A high test-retest reliability (0.80-0.95) of the different tests used in the present study was previously described (e.g., 2,27,28,38,39).

On the first day of each testing set, participants performed the following three tests:

Sprint Tests. Sprint ability was evaluated by a 10 m and 20 m standing-start all-out run with a 2-min rest period between runs. Time was recorded using photocell gates (Timing-Radio Controlled, TTSport, San Marino) placed 0.4 m above the ground, with an accuracy of 0.001s. The athletes performed three trials for each distance and the fastest times were recorded for further analysis. The runs were performed individually by each participant. The intra-class correlation coefficient for test-retest reliability and typical error of measurement for the 10 m and 20 m tests were 0.95, 0.97, and 1.3%, and 1.2%, respectively.

Agility Test. Agility was assessed by a handball agility specific test (HAST). This test included forward-backward running and diagonal sliding movements that resemble specific handball movements (39). The setup test is illustrated in Figure 2. Specifically, the player ran forward in a straight line until he reached a marking cone, 2, 5 m from the starting position, then he slid diagonally and backward to cone 3, which was positioned 3.5 m to the right of the starting position (cone 1). After the player slid diagonally and forward to cone 4, he ran backward in a straight line until he reached cone 5, placed 5 m from cone 4. Finally he slid diagonally and forwards to cone 3 and then slid diagonally and backwards to cone 1. For time measurement, this test was conducted using the same equipment as in the sprint tests. The faster of the two trials, with a 5-min recovery between trials, was recorded and used for statistical analysis. The intra-class correlation coefficient for test-retest reliability and typical error of measurement for the HAST was 0.92 and 2.3%, respectively.

FIGURE 2 ABOUT HERE

Upper Body Maximal Strength Test. Maximal strength of the upper extremity was assessed using one repetition concentric maximal bench press (1RMBP). The test was performed using a squatting apparatus in which the barbell was attached to both ends, with linear bearings on two vertical bars allowing only vertical movement (15). No bouncing or arching of the back was allowed. Warm-up consisted of a five-repetition set at a load of 50% of the perceived maximum weight. Thereafter, four to five separate single attempts were performed, starting from a weight of about 80% of the maximum perceived until the participant was unable to reach the full extension position of the arms. The last acceptable extension with the highest possible load was determined as 1RM. The rest period between attempts was always 2 min.

On the second day the participants performed the following two tests:

Vertical Jump Tests. Lower limb explosive power was assessed by a countermovement jump in two forms – with free arm-swing (CMJarm) and with hands on the hips (CMJ) (2). Participants began each jump in an erect standing position, and moved into a semi-squat position before jumping as high as possible. The athletes performed three trials for each jumping form with passive recovery of 60-s between jumps, and the best result was recorded. The height of each jump was estimated from the flight time by using the Optojump apparatus (Microgate Srl, Bolzano, Italy), and was expressed in cm. The intra-class correlation coefficient for test-retest reliability and typical error of measurement for the CMJ and CMJarm tests were 0.96, 0.97 and 2.7% and 2.9%, respectively.

YYIRTL1. The Yo-Yo intermittent recovery test level 1 (YYIRTL1) was used to assess players' aerobic capacity (38). The validity of the YYIRTL1 and its relevance for the assessment of intermittent high-intensity endurance in young male team handball players was previously described (38). The test consisted of 20-m shuttle runs performed at increasing velocities, with

10 sec of active recovery between runs until exhaustion. Recorded paces of the YYIRTL1 test were broadcast using a personal computer (500T1C-K03; Samsung Electronics Co., Ltd., Suwon, South Korea). The end of the test was determined when the player failed to arrive within 2m of the end line on two consecutive tones. The total distance (m) covered during the YYIRTL1 (including the last incomplete shuttle) was considered as the testing score.

Heart Rate Responses. HR responses were monitored during all the HIIT and SSG training sessions to provide the mean HR percentage (%HRmean) and the percentage of maximal HR (%HRmax) reached during training sessions. HR responses were recorded via a telemetry system (Hosand Technologies Srl, Verbania, Italy) at 5-sec intervals throughout each HIIT and SSG and during the YYIRTL1 (3). The maximum HR was determined as the peak HR observed during the YYIRTL1.

Statistical Analysis

A two-way repeated-measure analysis of variance was used to compare fitness differences, with time serving as the within-group factor and training regimen as the between-group factor. The Shapiro-Wilk test was used to ensure normal distribution of the results. Cohen's effect size (ES) statistic with 95% confidence intervals was used to determine the practical significance of observations (6). The ESs of, < 0.09, 0.10–0.49, 0.50–0.79, and > 0.80 were considered trivial, small, moderate, and large, respectively (6). The ICCs and typical error of measurement were used to determine the test-retest reliability of the dependent variables. Data are presented as the mean $\pm$ SD. Significance level was set at $p < 0.05$.

RESULTS

%HRmean during the training sessions in the SSGs training group ranged between $91.6 \pm 1.4\%$ and $93.1 \pm 1.1\%$ %HRmax, which was not significantly different ($p > 0.05$) than the corresponding HIIT group values ($90.4 \pm 1.1\%$ and $92.5 \pm 0.9\%$ HRmax). %HRmax reached during SSGs was $94.3 \pm 1.4\%$, similar to that recorded by HIIT group ($94.1 \pm 1.6\%$).

Baseline and the effect of training on anthropometric and fitness characteristics of the study subjects are summarized in Table 2 and Figure 3. There were no significant baseline anthropometric or fitness differences between the groups ($p > 0.05$). There was a significant improvement in the fat %, YYIRL1, 10m sprint, 20m sprint, HAST, 1RM bench press, CMJ, and CMJarm following training in both groups ($p < 0.05$, moderate to large effects).

There was a significantly greater improvement in the 10m sprint ($p < 0.05$, large effect), 20m sprint ($p < 0.05$, large effect), HAST ($p < 0.05$, large effect), 1RM ($p < 0.05$, large effect), CMJ ($p < 0.05$, moderate effect), and CMJarm ($p < 0.05$, large effect) following the SSGs training compared to the HIIT. No differences in aerobic capacity were found when comparing the effect of the two training methods ($p > 0.05$, small to trivial effect). No significant within- or between-groups differences were found in body weight following training ($p > 0.05$, trivial effect).

TABLE 2 ABOUT HERE

FIGURE 3 ABOUT HERE

DISCUSSION

The main purpose of this study was to compare the effects of HIIT versus SSGs training on selected fitness variables of elite male handball players during the in-season period. The results indicated that both the HIIT and the SSGs training program led to significant improvements in aerobic, anaerobic and strength performances (Table 2). SSGs training led to a greater improvement in the 10m and 20m sprint, CMJ, CMJarm, 1RM bench press, and HAST (Figure 3). No difference in aerobic capacity, assessed by the Yo-Yo intermittent running test, was found when comparing the effectiveness of the two training methods.

Data obtained through time-motion analysis has presented handball as a demanding sport, with numerous high-intensity actions taking place throughout the game (29,36). This suggests that the increase in the players' power and explosive capabilities should provide an advantage in sustaining the forceful muscle contractions required for activities such as hitting, blocking, pushing, holding, jumping, sprinting, and shooting during the game. Indeed, previous studies highlighted that power output of the leg extensor muscles, absolute jumping power, maximal upper-body strength, and sprint running are important neuromuscular performance characteristics for successful participation in elite levels of handball (16,28,36).

The results of the present study indicate that both SSGs training and HIIT were both efficient in improving the muscular maximal strength and power-related variables associated with handball performance (16,28,36). However, when comparing the effectiveness of the two, SSGs training was found to be significantly more efficient for improving power and strength compared to HIIT. The positive effect of SSGs on sprint and jump abilities are in accordance with previous studies reporting improved measures of 10m and 30m sprints and CMJ in soccer players (23,33,37). However, our results are in contrast to those of another longitudinal 10-week

study conducted in young handball players (3). In the latter no significant changes were found in the 10m sprint or the jumping performance following SSGs training and HIIT. As a possible explanation for this lack of training effect, the author reported that both training programs were specifically designed to improve aerobic rather than neuromuscular capacities. Nevertheless, the outcome of our study showed significantly better results of both sprint and vertical jump tests following SSGs training compared to HIIT training (Figure 3). One main reason for the advantage of the SSGs over the HIIT in improving lower limb power could be that during SSGs training players presumably perform higher amounts of high-to-maximal intensity actions than during HIIT (18,23,35). In fact, although not recorded in the present study, specific power-related actions, such as accelerations, decelerations, changes of directions, jumps, blocks, and stops, occur much more often during the SSGs training when compared to the HIIT. These actions involve intense muscular activity and high eccentric muscular efforts, which may have represented an efficient mean to increase lower leg muscular power (23,32). Outcomes from the present study have also revealed significant improved values of upper-body maximal strength in both groups following the training period. Yet, there was a significant “training type” interaction suggesting the superiority of SSGs training over HIIT to enhance these neuromuscular performance characteristics. Our results rely on the findings of a previous study (22) involving rugby player and aimed to investigate the physiological responses to an intensified period of rugby league competition and the subsequent impact on match performance. The author reported that the addition of physical contact to non-contact small-sided games results in upper body neuromuscular fatigue and marked and longer lasting increases in CK as well as increases in perceived effort compared to no physical contact. These results indicate that there is a significant amount of upper body fatigue induced by SSGs training that includes contacts. In addition, the

distinct movements performed during a match by the upper and lower body could contribute to the diversity in physiological responses and mechanisms of fatigue. For example, the lower body typically performs lower force, higher velocity movements during competition (e.g., running and jumping), whereas the upper body is involved in higher force, lower velocity movements (e.g., tackles, pushes, blocks and wrestles). Taken together, these results indicate that the high occurrence of actions such as hitting, blocking, pushing, and holding, and the greater frequency of physical contacts between players in the SSGs training, created a cumulative training stimulus that led to the significantly greater improvement in upper body strength, as demonstrated through the 1RM bench press, in the SSGs training over the HIIT group.

Another finding of the present study was the significantly greater improvement in agility after SSGs training compared to HIIT training. Agility is a complex ability that depends on several factors, such as muscle strength, balance, flexibility, and speed (8,23). Our results are in agreement with Katis and Kellis (23), who showed that 10 bouts of 4-min duration and 3-min passive recovery of three-a-side soccer SSGs training resulted in a significant improvement in the sprint and agility performance of young soccer players. The advantage of the SSGs training over the HIIT group in agility improvement in the present study may have resulted from the recurring “one-on-one” situations, which forced players to withstand and to overcome an opponent that tried either to score or to avoid goals. Obviously, such situations served as specific training drills for changes of direction and agility among the SSGs training group. In contrast, the HIIT group performed few changes of direction when involved in shuttle run HIIT training. As can be seen in Figure 1, players completed shuttle run HIIT training that included a change of direction during each running bout for a total of 24-32 per training session. Thus, in accordance with our hypotheses, it seems that SSGs training is a more efficient training method than HIIT

for agility enhancement. However, the fact that significant improvement in agility was also found in the HIIT group in the present study shows that the accelerations, decelerations, and high-speed running performed during this type of training is also sufficient to cause positive results for agility among handball players. Nevertheless, the use of three-a-side SSGs seems a preferred training method for handball players over HIIT, due to its higher positive impact on handball performance-related parameters.

Another main objective of the present study was to examine the effect of in-season HIIT and SSGs training on maximal aerobic power of the players. Since HIIT intensity can be controlled with much greater precision than that during handball games (5), we hypothesized that improvement would be greater for HIIT than for SSGs training. Our results revealed that both HIIT and SSGs training significantly increased the aerobic capacity of the players. However, in contrast to our hypothesis, both training methods were similarly effective in improving aerobic capacity. Short intermittent runs such as those designed in our training protocol are of particular interest for team sport coaches, as they enable athletes to improve aerobic as well as anaerobic performance capabilities (1,3,4,5,12,14,30). This particular form of exercise is characterized by the repetition of short running bouts at velocities close to maximal aerobic speed (MAS), alternating with short passive recovery periods. Similarly, the high frequency of high-intensity actions during SSGs training was found to be an effective alternative to the traditional intermittent training for improving aerobic fitness (3,4,18,19,24,31,34,35,40). Moreover, this training methodology maximizes the training time spent in a way similar in nature to that of handball match play. Although these two training methods were previously found to be effective in improving the physical fitness of young handball players (4,5), no study has compared their effectiveness among elite adult male handball players during the in-season period. In line with

our findings, Impellizzeri et al. (20) reported that 4 weeks of small-sided soccer game training during the pre-season was as efficient as high-intensity interval training for improving maximal oxygen uptake. Similarly, Dellal et al. (13) showed that 6 weeks of soccer SSGs training increase aerobic capacity and the ability to repeat high-intensity actions with directional changes of amateur soccer players at a proportion similar to that of the high-intensity intermittent exercise training. In our study the two training protocols were matched for exercise duration and recovery at each training session. Despite the different nature of the two training protocols, the physiological responses of the participants suggested that there was no significant difference between HIIT and SSGs training in terms of internal loads expressed by the %HRmean and peak %HRmax values recorded during training. These HR outcomes are consistent with previously reported similarly designed studies where HR ranges during SSGs training were between 82%-93% (3,4,5,10,12,17,18,19,24,34). Moreover the telemetric records of both HIIT and SSG training sessions in the present study were higher than those reported by Pavoas et al. (33) and Loftin et al. (26) concerning the physiological demands of handball players during official matches. Therefore, relying on the physiological responses recorded during the training sessions and referring to the optimal training regimes required to improve aerobic fitness (4,14,15), both prescribed protocols represented an efficient training stimuli to the aerobic fitness component. Thus, in contrast to our hypothesis, the current study revealed no significant differences between HIIT and SSGs training for the improvement of aerobic capacity. These results are in accordance with Buchheit et al. (3), who reported that 10 weeks of specific handball SSGs training produced equivalent improvements to high-intensity interval training on maximal aerobic power of junior handball players. These findings suggest that SSGs training and HIIT may have similar long-

term effects on the aerobic capacity of elite handball players. Thus, in order to progressively develop aerobic capabilities, handball coaches could use both methodologies indiscriminately.

PRACTICAL APPLICATIONS

The present study demonstrated that an 8-week period intervention, including either HIIT or SSGs training sessions twice a week, could improve the fitness performances of elite adult handball players during the in-season period.

HIIT and SSGs training were equally effective in developing aerobic capabilities and supra-maximal intermittent performance (according to the YYIRL1).

SSGs training, applied during 8 weeks, achieved greater improvement of anaerobic and strength performances assessed by 10 and 20m sprints, vertical jumps, agility, and 1RM bench press test, compared to HIIT. Consequently, during the in-season period, handball coaches may prefer the SSGs training instead of HIIT to optimize training.

Finally, coaches and players could simultaneously solicit higher physical demands than those presented in an official match-play (maximal aerobic power, changes of directions, high-intensity actions) in order to continually develop handball players' fitness components while also encompassing handball-specific technical and tactical elements.

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ACKNOWLEDGMENTS

The authors wish to thank the Maccabi Tel Aviv Handball Club for its cooperation in the study. The authors also wish to thank each of the players and the coaches for their efforts in completing the physical tasks needed for the study.

Figure Legends

FIGURE 1. An example of an intermittent run patterns for a participant with a Yo-Yo maximal speed (YYIRTL1) of 16 km/h^{-1} . For high intensity intermittent running (HIIT) the participant runs within 15s in the form of a shuttle run between an inner and an outer cone.

FIGURE 2: Handball agility specific test (HAST) setup.

FIGURE 3: Improvement in fitness characteristics following the different training protocols (* $p < 0.05$ for between group differences).

TABLE 1. Description of the training schedule and details of training protocol over the 8-week training period in high intensity intermittent training (HIIT) and small-sided games (SSG).

		HIIT	SSG
		Sprint Tests (10m, 20m), Agility Test (HAST), Maximal Strength and Lower Limb Explosive Power, Yo-Yo IR1	
Testing	Week-1	Sprint Tests (10m, 20m), Agility Test (HAST), Maximal Strength and Lower Limb Explosive Power, Yo-Yo IR1	
	Week-2	2 x [2 x 6':15"(90%)-15"p]	2 x [5 x 2'25"-1'p]
Training Period	Week-3	2 x [2 x 6'30":15"(90%)-15"p]	2 x [5 x 2'35"-1'p]
	Week-4	2 x [2 x 7':15"(92%)-15"p]	2 x [5 x 2'55"-1'p]
	Week-5	2 x [2 x 7'30":15(92%)-15"p]	2 x [5 x 3'-1'p]
	Week-6	2 x [2 x 7'30":15(92%)-15"p]	2 x [5 x 3'-1'p]
	Week-7	2 x [2 x 8':15(92%)-15"p]	2 x [5 x 3'10"-1'p]
	Week-8	2 x [2 x 7'30":15(95%)-15"p]	2 x [5 x 3'-1'p]
	Week-9	2 x [2 x 7':15(95%)-15"p]	2 x [5 x 2'55"-1'p]
Testing	Week-10	Sprint Tests (10m, 20m), Agility Test (HAST), Maximal Strength and Lower Limb Explosive Power, Yo-Yo IR1	

As for HIIT protocol, 2 x [2 x 6':15"(90%)-15"p] in week-1 means: 2 sessions per week consisting of 15-s runs at 90% of VYoYoIR1 interspersed with 15-s (15"p) passive recovery by walking for a total time of 6'15". As for SSG protocol, 2 x [5 x 2'25"-1'p] in week-1 means: 2 sessions per week consisting of 5 bouts of 2'25" of continuous handball small sided games with 1' (1'p) of passive recovery between bouts.

TABLE 2. The effect of both training methods on anthropometric and fitness characteristics of the participants (mean $\pm$ SD).

	High Intensity Training (n=9)		Small-Sided Games (n=9)	
	Pre	Post	Pre	Post
Weight (kg)	91.0 $\pm$ 8.5	91.1 $\pm$ 7.5	90.1 $\pm$ 10.1	90.7 $\pm$ 9.8
Height (cm)	185.9 $\pm$ 5.0	185.9 $\pm$ 5.0	188.8 $\pm$ 5.7	188.8 $\pm$ 5.7
Body Fat (%)	13.3 $\pm$ 3.7	12.8 $\pm$ 2.8*	12.0 $\pm$ 2.6	11.3 $\pm$ 2.1*
YYIRL1 (m)	1297.8 $\pm$ 300	1601.1 $\pm$ 192*	1364.4 $\pm$ 397	1723.3 $\pm$ 327*
10m Sprint (sec)	1.55 $\pm$ 0.08	1.52 $\pm$ 0.07* [†]	1.54 $\pm$ 0.12	1.48 $\pm$ 0.12*
20m Sprint (sec)	2.80 $\pm$ 0.10	2.75 $\pm$ 0.12* [†]	2.81 $\pm$ 0.12	2.70 $\pm$ 0.10*
Agility Test (sec)	6.72 $\pm$ 0.22	6.65 $\pm$ 0.24* [†]	6.68 $\pm$ 0.25	6.54 $\pm$ 0.21*
Bench Press (kg)	99.4 $\pm$ 10.1	106.2 $\pm$ 10.7* [†]	105.0 $\pm$ 22.6	118.2 $\pm$ 21.0*
CMJ (cm)	34.3 $\pm$ 5.0	36.9 $\pm$ 4.5* [†]	36.5 $\pm$ 4.5	40.5 $\pm$ 4.5*
CMJarm (cm)	40.2 $\pm$ 5.9	42.8 $\pm$ 5.3* [†]	42.6 $\pm$ 5.8	46.4 $\pm$ 5.2*

* p<0.05 for within group changes.

[†] p<0.05 for between group changes.

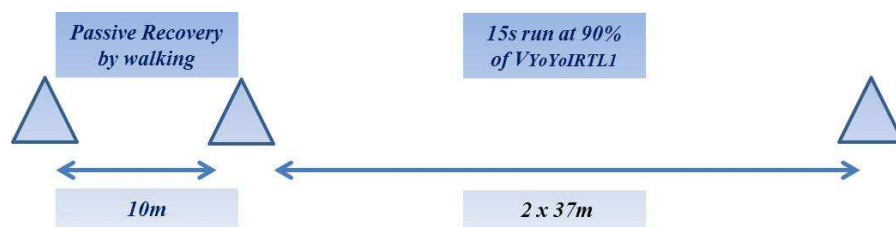


Figure 1. An example of an intermittent run patterns for a participant with a Yo-Yo maximal speed (YYIRTL1) of 16 km/h^{-1} . For high intensity intermittent running (HIIT) the participant runs within 15 s in the form of a shuttle run between an inner and an outer cone.

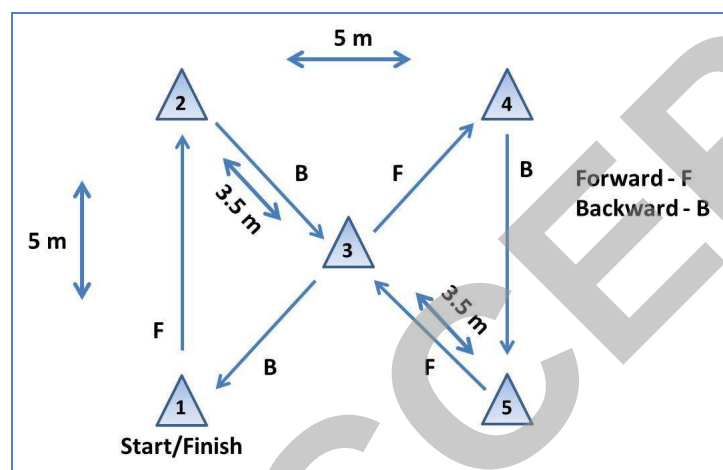


Figure 2. Handball agility specific test (HAST) setup.

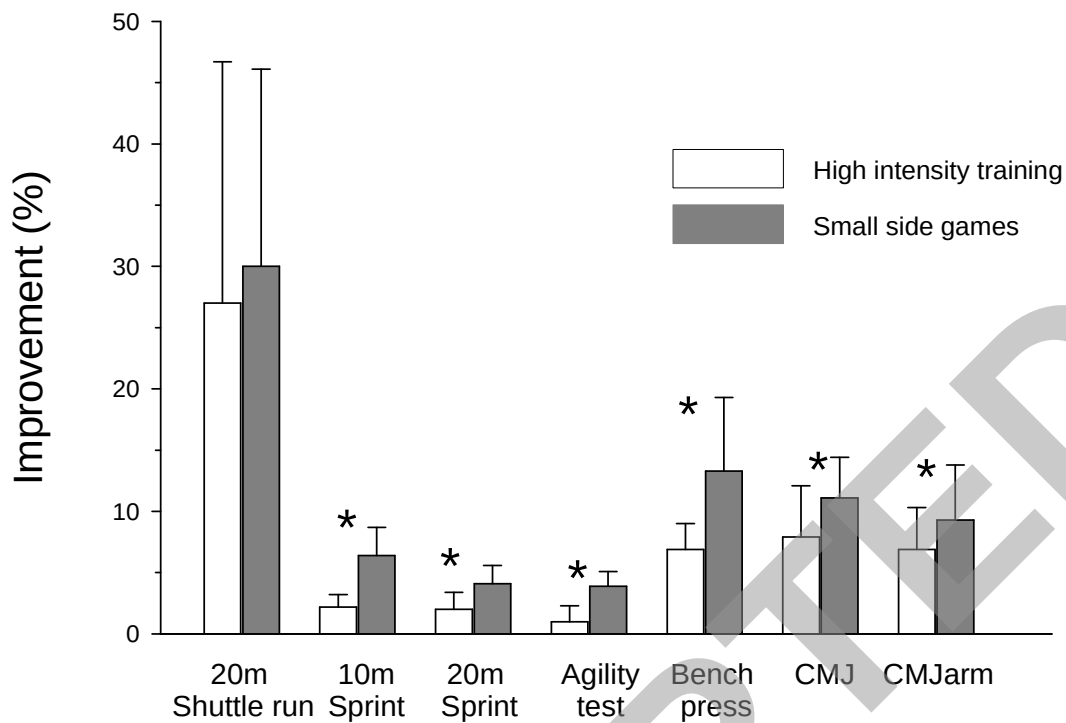


Figure 3. Improvement in fitness characteristics following the different training protocols (* $p < 0.05$ for between group differences).